

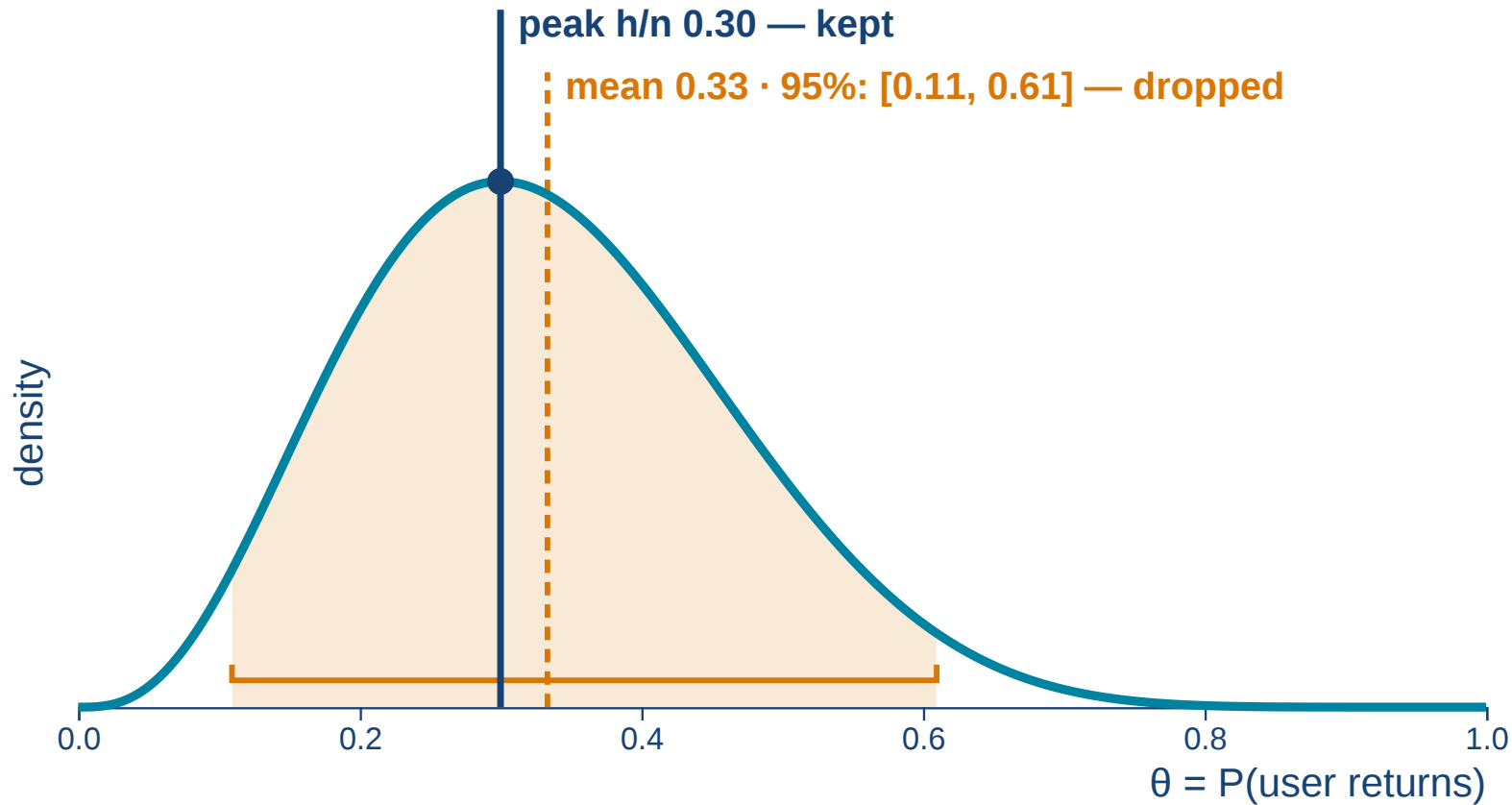
Probabilistic Machine Learning

Lecture 03: Keep the Peak, the Generative Route and Naive Bayes

Prof. Dr. Michael Granitzer

Where We Left Off

Lecture 02: 3 of 10 users, flat prior $\rightarrow$ Beta(4, 8)



One posterior, three readings:

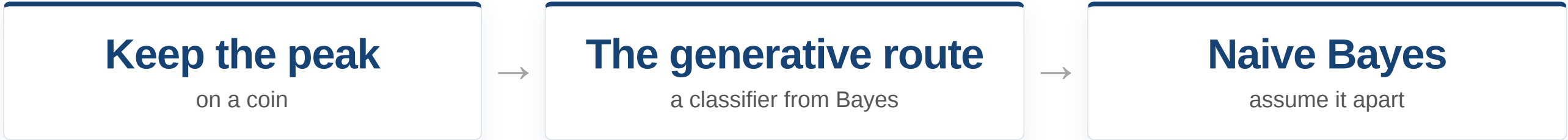
- **peak** $0.30 = h/n$
- **mean** 0.33
- **95% interval** [0.11, 0.61]

Today keeps only the **peak** — and then asks what a classifier built on probability looks like.

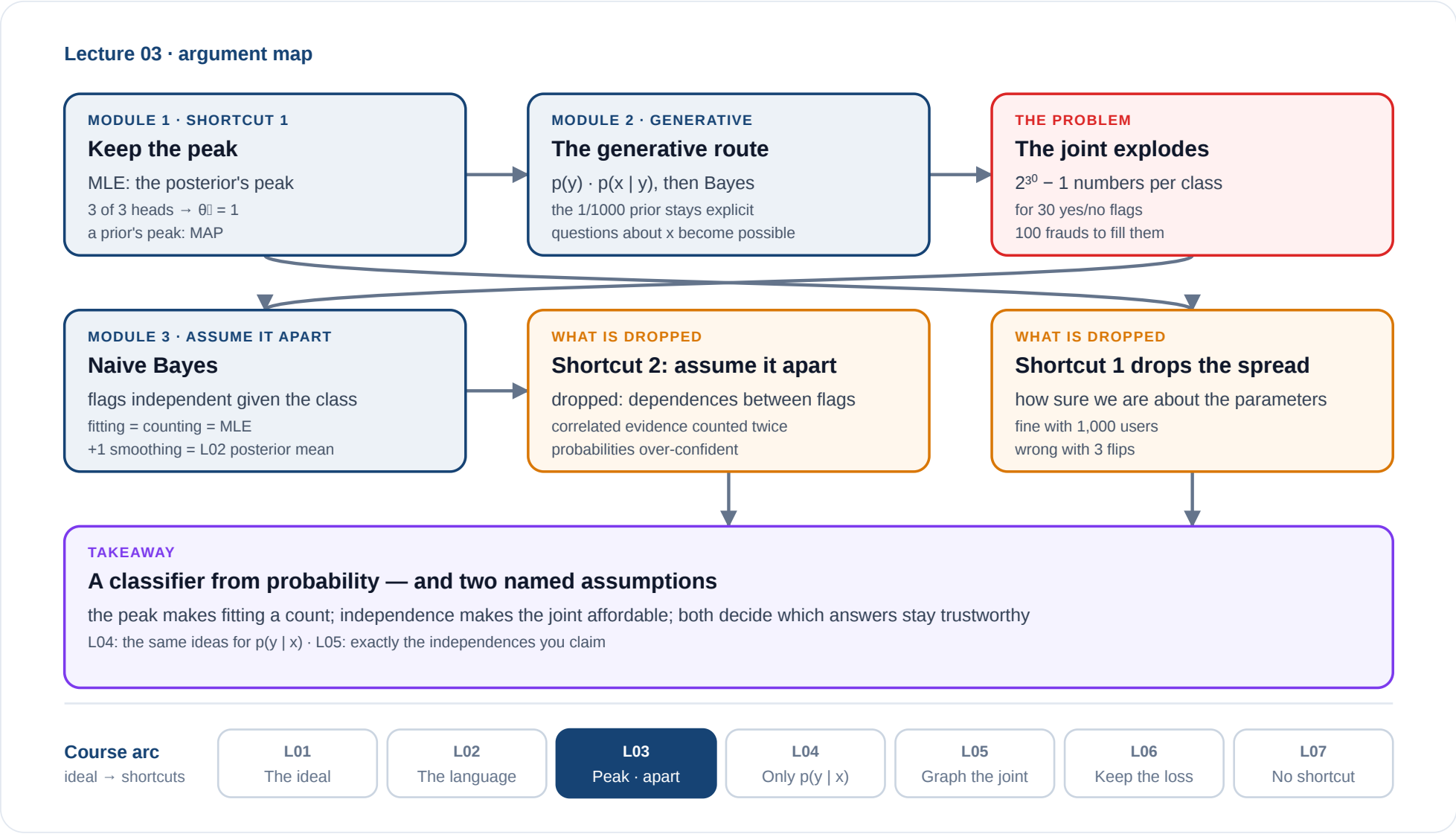
Today: Keep the Peak, Assume It Apart

Lecture 01's table, rows 1 and 3 — and the first classifier built straight from probability:

shortcut	instead of the ideal	what is lost	where
Keep the peak	the most probable parameter — MLE or MAP	how sure we are about it	a coin · every count
Assume it apart	factorise the joint by independence	the dependences assumed away	naive Bayes



Argument Map



Keep the Peak

Guiding Question: Last week's posterior gave a peak, a mean and an interval. What do we lose when we keep only the peak — and why does almost everyone do it?

The Likelihood, Read Along the Other Axis

The **likelihood** is the probability of the observed data, read as a function of the **parameter** with the data held fixed: $L(\theta) = p(D \mid \theta)$. The **maximum likelihood estimate** is its peak, $\hat{\theta}_{\text{ML}} = \arg \max_{\theta} p(D \mid \theta)$.

Lecture 02, 3 of 10 users

$$p(\theta \mid D) \propto \theta^3(1 - \theta)^7 \cdot 1$$

With a **flat prior** the posterior *is* the normalised likelihood — so its peak $h/n = 0.3$ is the maximum likelihood estimate.

Not a distribution over θ

$$\int_0^1 \theta^3(1 - \theta)^7 d\theta = \frac{1}{1320}$$

The likelihood is a probability over **data**. Read along θ it need not integrate to 1 — only the posterior does.

Deriving h/n Without a Posterior

Lecture 02's i.i.d. assumption lets the likelihood factorise over rows; the logarithm turns the product into a sum without moving the peak:

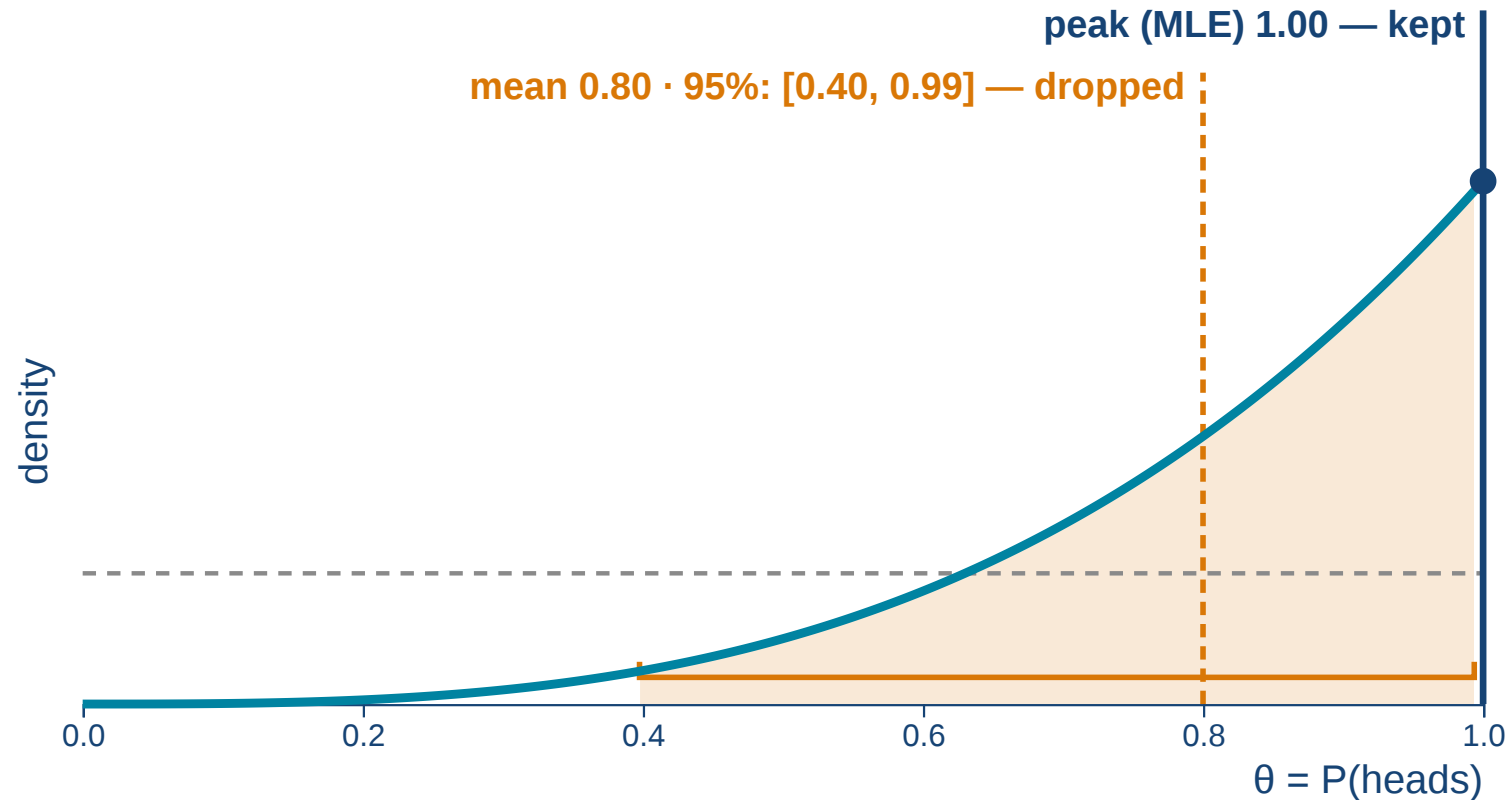
$$\ell(\theta) = \log \prod_{i=1}^n \theta^{y_i} (1 - \theta)^{1-y_i} = h \log \theta + (n - h) \log(1 - \theta)$$
$$\frac{d\ell}{d\theta} = \frac{h}{\theta} - \frac{n - h}{1 - \theta} = 0 \quad \implies \quad \hat{\theta}_{\text{ML}} = \frac{h}{n}$$

Last week's peak, found by **setting a derivative to zero** — no prior, no normaliser, no integral. The recipe works unchanged for a million parameters, where nobody can draw the posterior.

The Peak Can Lie

Three flips, **three heads**. Maximum likelihood: $\hat{\theta}_{\text{ML}} = 3/3 = 1$.

Flat prior (dashed), 3 heads: posterior Beta(4, 1)



Predict before the second click: where do the posterior mean and the 95% interval lie?

A Prior's Peak: MAP

Keep the prior, and still take only the peak — the **maximum a posteriori** estimate. With a $\text{Beta}(a, b)$ prior, the same derivative gives

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} p(D \mid \theta)p(\theta) = \frac{h + a - 1}{n + a + b - 2}$$

prior	data	MLE h/n	MAP	posterior mean
Beta(1, 1) flat	3 of 3	1	1	0.80
Beta(2, 2)	3 of 3	1	0.80	0.71
Beta(2, 2)	8 of 10 (Lecture 02)	0.80	0.75	0.71

The prior shifts the peak as if $a - 1$ yeses and $b - 1$ noes had been seen before the data — a **flat** prior shifts nothing. MAP pulls 3 of 3 back to 0.8 — and still reports one number.

Why Everyone Keeps the Peak

	the whole posterior	only the peak
needs	the normaliser $p(D)$ — an integral over θ	a derivative
in 1 dimension	a Beta table	h/n
in a million	an integral nobody can compute exactly	gradient descent
costs	computation	the spread

With **1,000 users** the posterior is Beta(301, 701): standard deviation 0.015. The peak is nearly everything there is to know.

Mini-Exercise

Two analysts both report $\hat{\theta} = 0.3$. One saw **3 of 10** users return, the other **300 of 1,000**.

What does the reported number hide — and which estimate would you bet on?

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The **spread**. Lecture 02's posteriors: $[0.11, 0.61]$ after 10 users, $[0.27, 0.33]$ after 1,000 — the same peak, an interval nine times wider. Bet on the second; a point estimate cannot tell you to.

Takeaway

Shortcut 1 — keep the peak. Maximum likelihood (flat prior) and MAP (with a prior) keep the posterior's most probable point. It is cheap — a derivative instead of an integral — and harmless with plenty of data. It drops the **spread**: how sure we are about the parameter.

The Generative Route

Guiding Question: A classifier needs $P(\text{fraud} \mid x)$. Bayes' theorem builds it from how fraud *looks* — what would it take to model that, and why does nobody do it in full?

Two Routes to the Same Posterior

A labelled table of transactions supports **both** routes to $P(\text{fraud} \mid x)$, where x is the vector of red flags:

DISCRIMINATIVE — LECTURE 04

Model $p(y \mid x)$ directly

A score for each x , as logistic regression will give (Lecture 04). Nothing is said about how x arises.

GENERATIVE — TODAY

Model $p(y)$ and $p(x \mid y)$

How does a *fraudulent* transaction look, and how an honest one? Then convert with Bayes.

$$P(y \mid x) = \frac{p(x \mid y) \cdot P(y)}{\sum_{y'} p(x \mid y') P(y')}$$

Why Model How Fraud Looks?

A discriminative model — logistic regression, the next lecture — gives $P(\text{fraud} \mid x)$ directly. What can a model of $p(x \mid y)$ do that it cannot?

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- ✓ **Keep the prior explicit.** The 1 in 1,000 of Lecture 02 is a separate factor: a new market with 1 in 200 changes $P(y)$, not the whole model.
- ✓ **Handle a missing field.** An attribute nobody observed can be summed out of $p(x \mid y)$.
- ✓ **Notice the strange.** A transaction with tiny $p(x)$ under *both* classes is unlike anything seen — a score alone cannot say so.

The price: $p(x \mid y)$ is a **joint distribution over all attributes**, one per class.

The Joint Explodes

30

yes/no red flags per transaction



$2^{30} - 1$

≈ 1.07 billion numbers for $p(x \mid \text{fraud})$ alone



100

frauds among 100,000 transactions to estimate them

Almost every cell of the table is **empty**. Its count is 0, so its maximum likelihood estimate is 0 — every fraud that looks slightly new becomes "impossible".

Mini-Exercise

A fraud model uses only **three** yes/no flags.

How many numbers does the generative model need — both classes and the class prior together?

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15. Each class needs a full table over $2^3 = 8$ flag patterns: 7 free numbers (they sum to 1). Two classes give 14, and $P(\text{fraud})$ adds one.

Takeaway

The generative route models **how the data arises** — $p(y)$ and $p(x | y)$ — and converts with Bayes. It keeps the prior explicit and can answer questions about x itself, but it must pay for a **joint over all attributes**, which explodes.

Naive Bayes: Assume It Apart

Guiding Question: Its central assumption is almost always false, and it classifies well anyway. How can both be true?

Conditional Independence

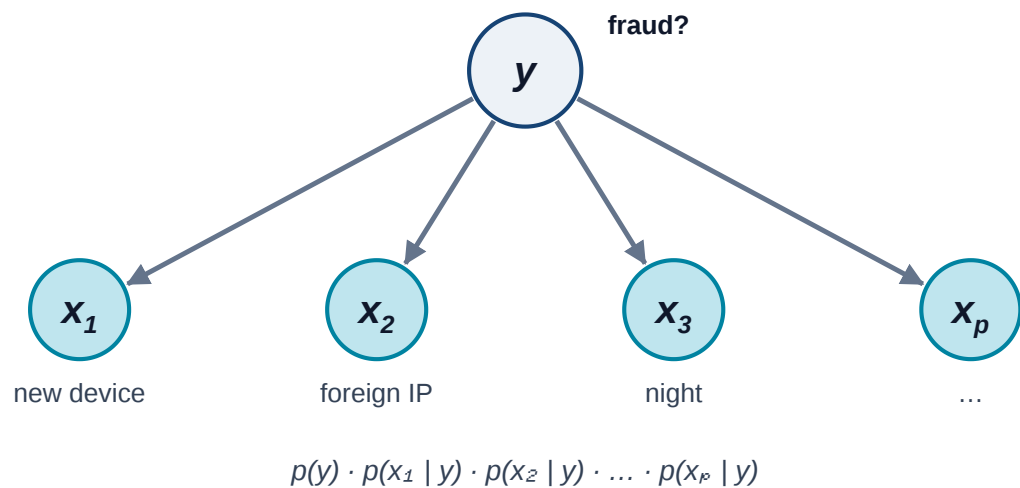
Definition. A and B are **conditionally independent given C** , written $A \perp\!\!\!\perp B \mid C$, iff

$$P(A \cap B \mid C) = P(A \mid C) \cdot P(B \mid C) \iff P(A \mid B \cap C) = P(A \mid C)$$

Reading: **once C is known, B tells us nothing more about A .**

Among frauds only: does a new device make a foreign IP more likely?

The Naive Assumption



Assume the attributes are **independent given the class**:

$$p(x_1, \dots, x_p \mid y) = \prod_{j=1}^p p(x_j \mid y)$$

flags	full joint, per class	naive
3	7	3
30	1,073,741,823	30

Shortcut 3 — assume it apart. Dropped: every dependence *within* a class.

Fitting Is Counting

$$\hat{P}(y) = \frac{n(y)}{n} \quad \hat{P}(x_j = v \mid y) = \frac{n(x_j = v, y)}{n(y)}$$

Lecture 02

Each attribute, within one class, is a **categorical** variable — for a yes/no flag, a Bernoulli.

Module 1 today

Its maximum likelihood estimate is the **relative frequency** — the same derivation as h/n for the coin.

Naive Bayes is fitted in one pass over the table, with no optimiser — and every number it uses is a **peak**. Shortcut 1 is taken as well.

The Classifier

$$P(y \mid x_1, \dots, x_p) = \frac{P(y) \prod_j p(x_j \mid y)}{\sum_{y'} P(y') \prod_j p(x_j \mid y')}$$

For two classes, divide the two posteriors — the normaliser cancels:

$$\underbrace{\frac{P(\text{fraud} \mid x)}{P(\text{honest} \mid x)}}_{\text{posterior odds}} = \underbrace{\frac{P(\text{fraud})}{P(\text{honest})}}_{\text{prior odds}} \cdot \prod_{j=1}^p \underbrace{\frac{p(x_j \mid \text{fraud})}{p(x_j \mid \text{honest})}}_{\text{likelihood ratio LR}_j}$$

Every observed attribute **multiplies the odds** by its likelihood ratio: above 1 it speaks for fraud, below 1 against.

Three Red Flags

100,000 transactions, 100 of them fraud: prior odds 1 : 999.

flag	among frauds	among honest	likelihood ratio
new device	60%	10%	6
foreign IP	70%	5%	14
night	50%	20%	2.5

A transaction shows all three flags. Is it more likely fraud than not?

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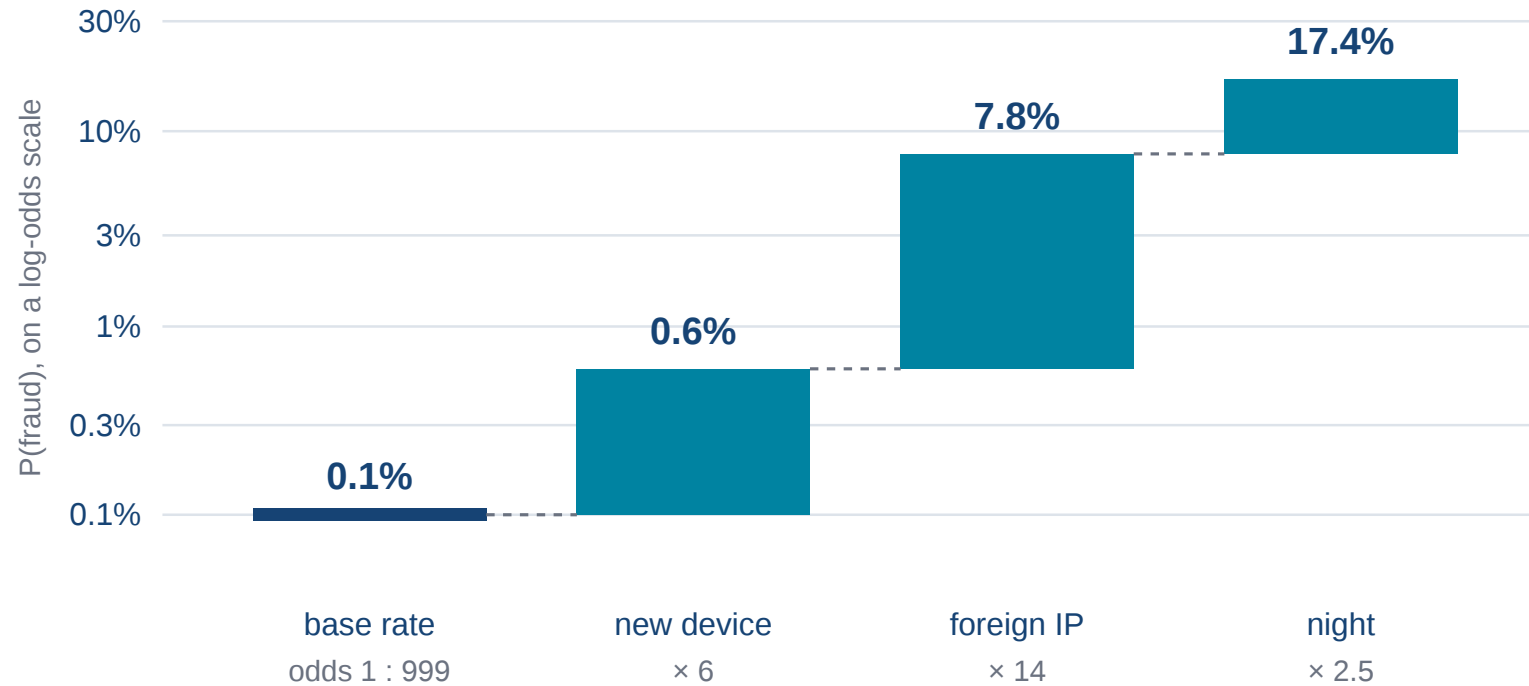
A transaction shows all three flags. Is it more likely fraud than not?

$$\text{odds} = \frac{1}{999} \cdot 6 \cdot 14 \cdot 2.5 = \frac{210}{999} \implies P(\text{fraud} \mid x) = \frac{210}{1,209} \approx 17\%$$

Three red flags, each one strong — and still **mostly honest**. The base rate of Lecture 02, once more.

Evidence Adds Up

odds 210 : 999 $\rightarrow$ $P(\text{fraud}) = 17.4\%$



Predict before each click: which flag moves the probability most?

One Zero Vetoes Everything

A new flag: **payment to a crypto exchange**. Among the 100 frauds it never occurred.

$$\hat{P}(\text{crypto} \mid \text{fraud}) = \frac{0}{100} = 0$$

A transaction with this flag gets posterior odds **exactly 0** — whatever the other thirty flags say. One unseen value vetoes all other evidence.

KEEP THE PEAK, FAILS

"Not seen in 100 frauds" became "impossible for a fraud" — the three heads in three flips of Module 1, in a single cell.

Smoothing Is a Prior

Laplace smoothing adds one pseudo-count to every value of attribute j (k_j values):

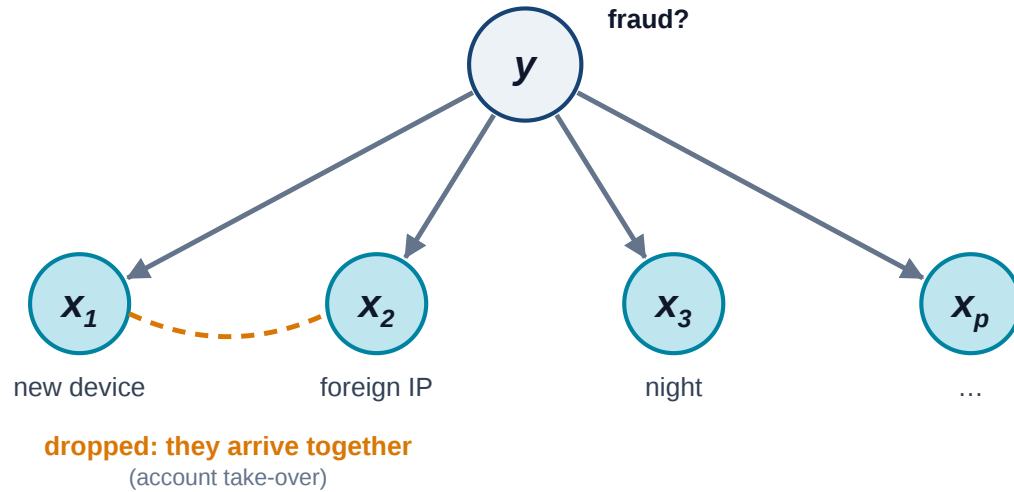
$$\hat{P}(x_j = v \mid y) = \frac{n(x_j = v, y) + 1}{n(y) + k_j}$$

For a yes/no flag ($k_j = 2$) this is $\frac{h+1}{n+2}$ — the **posterior mean** $\frac{a+h}{a+b+n}$ of Lecture 02 under the flat prior Beta(1, 1). For $k_j > 2$ values, the flat Dirichlet does the same.

count among 100 frauds	MLE (peak)	smoothed (mean)
0	0	$1/102 \approx 0.010$
60	0.60	$61/102 \approx 0.598$

Under the flat prior the peak of 0 of 100 is still 0. Smoothing quietly **leaves the peak for the mean** — exactly where the peak was wrong.

Counting Evidence Twice



In an account take-over, a new device and a foreign IP **arrive together**. Suppose both appear on 55% of frauds and 4% of honest transactions:

	ratio for the pair	$P(\text{fraud})$ with night
naive Bayes	$6 \cdot 14 = 84$	17.4%
the real pair	$0.55/0.04 \approx 14$	3.3%

Correlated flags are **counted twice**, so naive Bayes is **over-confident**. Its ranking of transactions often survives; its probabilities do not.

Mini-Exercise

A bug feeds the **foreign-IP** flag into the model twice.

Does the fraud probability of our three-flag transaction go up, down, or stay — and what does that say about correlated attributes?

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Up — from 17% to about 75%. The odds get a second factor 14: 2,940 : 999. Naive Bayes cannot tell a copy from new evidence; a strongly correlated attribute is almost a copy.

Takeaway

Naive Bayes **assumes the joint apart** — attributes independent given the class — and **keeps the peak** of every count; smoothed, it keeps the Beta posterior *mean* instead. It is cheap, needs little data and ranks well. Its probabilities are over-confident wherever attributes move together.